

This question focuses on estimating causal effects. This means that instead of just looking for correlation, we are looking for causation, which often requires more in-depth modeling and assumptions.

There are two main parameters of interest when estimating causal effects. One is called the ATE. This is the average treatment effect for the population. When we run a simple linear regression model on the outcome and treatment variables (in this dataset outcome is `bwght` and the treatment is `smoking_cigs`) for the whole population, the β_1 coefficient can be interpreted as the average difference in `bwght` between smokers and non-smokers. Under the right assumptions, this can estimate the ATE.

The other parameter of interest is called the ATT. This is the average treatment effect for the treated. It focuses on the treated group and tries to estimate what their `bwght` might have been had they not smoked, compared to what it actually is.

While both the ATE and the ATT are useful for estimating causal effects, they each have scenarios where they are more useful than the other. The ATT can more accurately estimate causal effects in non-experimental settings where the treated and non-treated groups are very different. This is because the differences in characteristics between the treated and control groups can affect both the treatment and outcome likelihood. This is the case with our data. As you can see from the table in questions 3 and 4, the characteristics of the treated (`smoking_cigs`) and non-treated (`not smoking_cigs`) groups are very different. On average, the non-smokers have a higher income and parental education. While controlling for these characteristics using multiple linear regression can help reduce some bias, estimating the ATT is an alternative and sometimes more effective way to do this. This is what this question explores.

Note: Although we just use the `estimate_att` function once in the problem set, it can be used multiple times in practice to impose the restriction of overlap (subsetting the data to make the characteristics of the treated and non-treated more similar) to varying degrees and to run different regression models.